



Optikinesis

Hands-free computer access for people who cannot rely on their hands.

Webcam-based cursor + blink selection + voice shortcuts · optional IMU
MotionGlass for precision

Samsung Solve for Tomorrow 2026

Theme : Health & Education

Team Revolutionary

Aditya Kumar · Govind Upadhyay · Himal Garg

Core thesis

Digital independence should not require expensive specialist hardware.

Start with software. Upgrade only when a user needs higher precision.



TEAM & OUR STORY

ORIGIN STORY

During our Community Engagement course, we visited an NGO and met people living with ALS / MND-related mobility limitations.

Their mind was present but an interface was missing

Everyday digital actions still depended on a caregiver

Can we restore everyday digital independence using the hardware people already have?

That question became Optikensis: webcam-first access today, MotionGlass precision for users who need more stability.

Individual contribution map

Himal Garg

Product & AI/CV Lead

Problem framing, MediaPipe/OpenCV head-pose control, blink-to-click logic, system architecture, prototype demo and pitch narrative.

Aditya kumar

Hardware & IoT Lead

MotionGlass concept, IMU + BLE architecture, sensor selection, wearable prototyping plan, device diagnostics and hardware feasibility.

Govind Upadhyay

Research, UX & Impact Lead

NGO insight documentation, user journey, accessibility workflow design, testing plan, caregiver feedback and pilot-partner outreach.

“Digital independence should not be limited by whether a patient can afford specialist assistive hardware.”

PROBLEM

2.5 Billion+

people globally need at least one assistive product today and is projected to exceed 3.5B by 2050.

1 Billion

people who need assistive technology are denied access, especially in low- and middle-income settings.

5.4 Million

Indians were recorded with movement disability in Census 2011; 26.8M people with disability overall.

3%

access to needed assistive products can be as low as 3% in some countries.

AFFECTS PATIENT PHYSICALLY , EMOTIONALLY AND MENTALLY

ALS patient undergoes court-approved euthanasia at Sheba hospital: 'My life now is hell and I want to end it'

The 44-year-old man who described himself as a 'bag of bones' made his plea before the loss of his last ability to communicate by blinking his eyes; His oxygen was reduced to the same level found in ambient air and was given palliative sedation to prevent any sensation of suffocation, which complies with the Patients' Rights Law

Higher Suicide Risk Found for US Veterans With ALS

Written by [Forest Ray PhD](#) | Feb. 22, 2021

Sara Bennett, influencer with ALS, shares final message while announcing her death at 39 - 'I am not in pain, or...'

EXPENSIVE AND INACCESSIBLE EXITING SYSTEMS

PRESS RELEASE

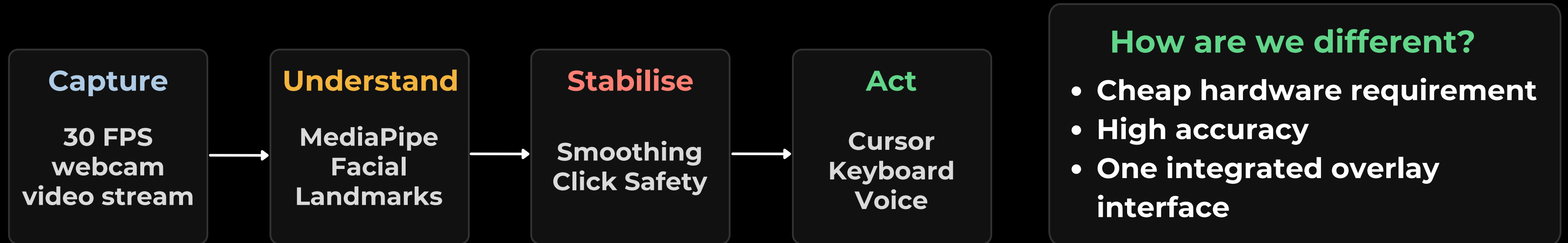
ALS Care Costs Over \$47,000 in the First Year, ALS Association-Led Study Shows

Out-of-pocket ALS costs create heavy financial burden for French families

Study: Patients, caregivers often need to stop working to manage disease

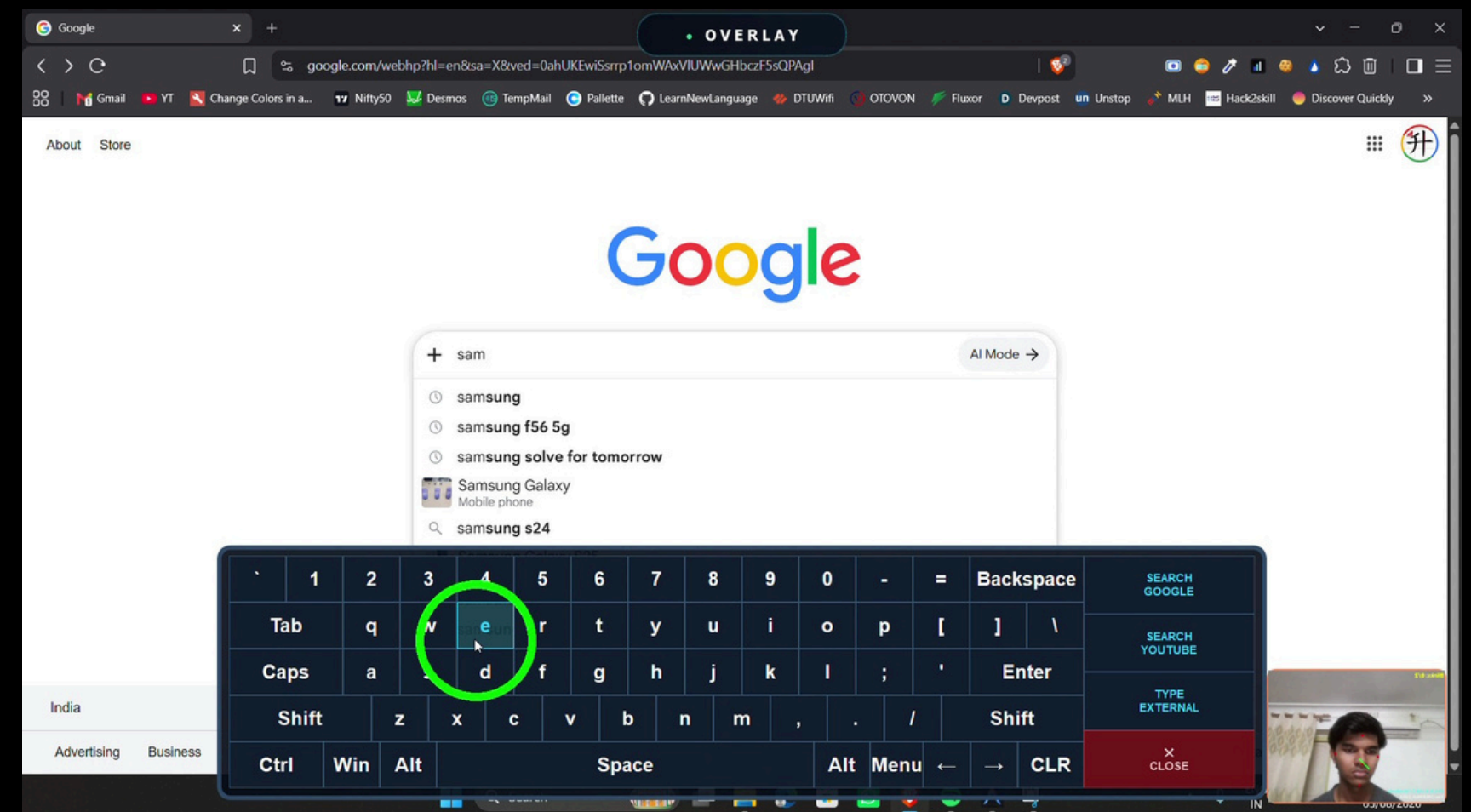
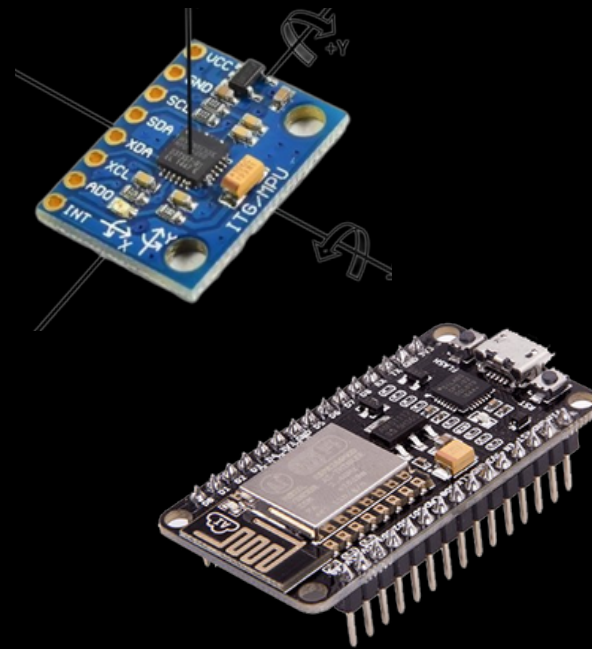
SOLUTION

OptiKinesis: A hands-free control system from an ordinary webcam.



Upcoming prototype: MotionGlass

A lightweight IMU-enabled wearable that integrates with OptiKinesis to deliver smoother, faster, and more precise hands-free computer control. By combining inertial sensing with computer vision.



CUSTOMER

TYPICAL CUSTOMER



SAKSHI VERMA

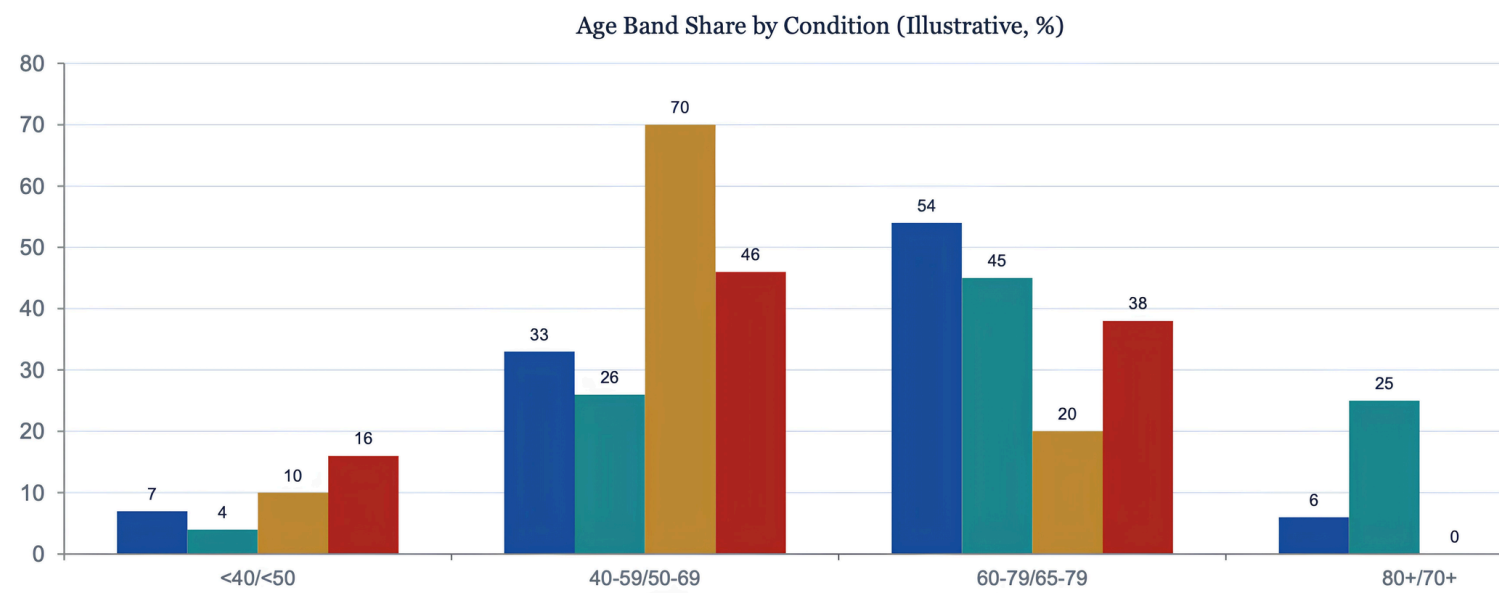
- Age: 47
- Occupation: Senior Accountant
- Condition: Early-stage ALS
- Devices: Laptop, Webcam
- Location: Urban, India

A working professional (40–60 years) experiencing progressive upper-limb motor impairment who seeks to remain independent at work and in daily life.

MARKET INSIGHTS

AGE WISE CUSTOMER SPLIT

Every condition skews older — except CTS



DISTINCTIVE BEHAVIOUR



Still in workforce high motivation to remain productive



Prefers software that works with existing laptop rather than dedicated hardware

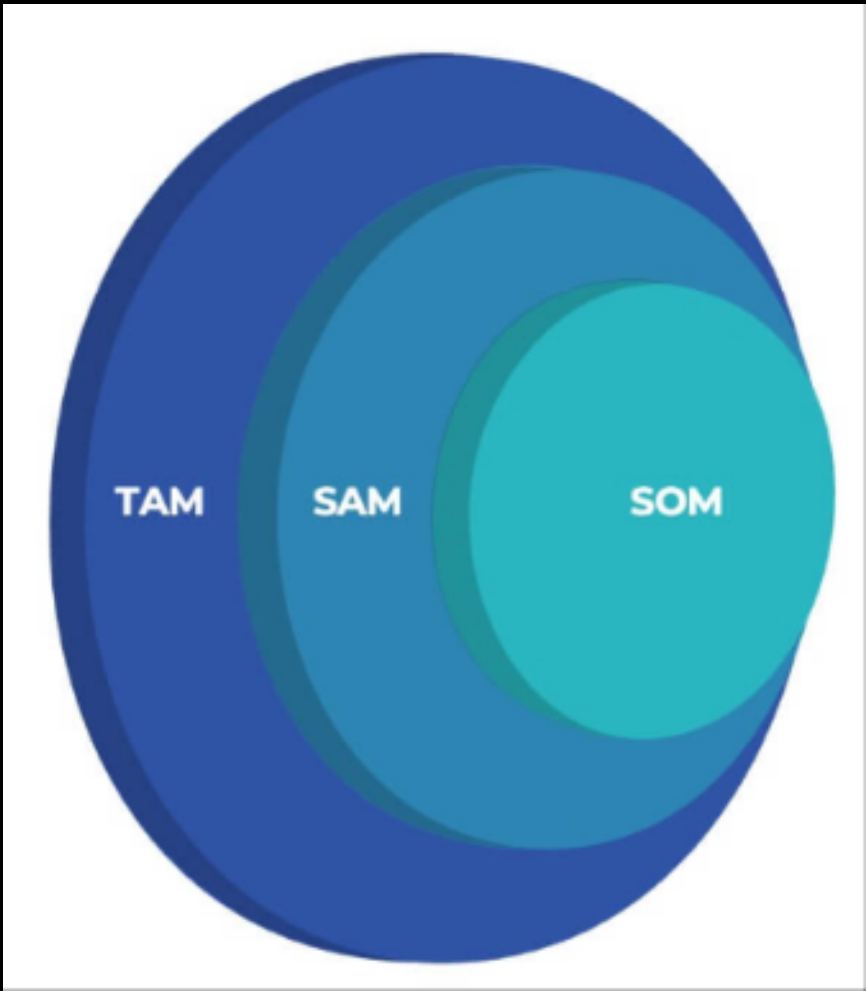
HOW WE CLASSIFY THEM



By Condition
ALS/MND (primary), Parkinson's, Stroke, CTS



By Stage
Early to mid-stage motor impairment



TAM Total Addressable Market

\$2.8B

Global AAC Device Market in 2026
Growing to \$5–6B by 2035 (9–11% CAGR)

SAM Serviceable Available Market

75M+

People globally require assistive technology for daily activities (WHO)

SOM Serviceable Obtainable Market

1–2M

Initial focus on ALS & Parkinson's patients via hospitals, rehab centres, NGOs & direct-to-consumer

Competition / Unique Value Proposition

Product	Access Method	List Price	Where It's Weak
 Tobii Dynavox TD I-Series/TD Pilot	Dedicated eye tracker + SGD software	\$10,000–\$20,000	Expensive; requires insurance/funding approval; proprietary hardware
 Eyegaze Edge (LC Technologies)	Dedicated eye tracker	~\$8,000–\$15,000	Similar cost barrier; US-centric distribution
 Tobii PCEye + tablet/laptop	Add-on eye tracker on user hardware	~\$1,200–\$2,000	Cheaper but still hardware-dependent; no emergency layer
 Windows Eye Control/ macOS Switch Control	Built-in OS accessibility	Free (needs compatible hardware)	Basic; no blink SOS; no caregiver auto-alert; limited customization
 OptiKinesis (proposed)	Standard webcam — gaze + blink, no dedicated hardware	Target: low-cost/ subscription (TBD)	No clinical validation yet; unproven accuracy & reliability at scale

What is unique that others don't have

- Software-first + hardware-upgrade path
- Very low hardware dependency
- All-in-one accessibility ecosystem
- Emergency support + caregiver controls

How are we better

- Works with a normal webcam, so core access does not need special hardware.
- Cheaper entry point because the software can start first.
- Gives more than just pointing: blink click, voice, virtual keyboard and SOS help.
- Optional MotionGlass can improve precision later without replacing the whole system.

What is common in this market

- Users want something easy to start and easy to learn.
- Families prefer low cost and low hardware dependency.
- Hospitals and NGOs want solutions they can deploy at scale.

BUSINESS MODEL

Affordable entry, sustainable support, responsible upgrades.

Access Software one-time purchase

webcam control, blink click, virtual keyboard,
voice shortcuts, SOS, local diagnostics

Optikensis Care optional support

quarterly calibration, device-health check,
priority support, caregiver backup

MotionGlass Kit target hardware kit

IMU wearable, Bluetooth setup, Motion/Fusion
Mode, charger, setup guide

Institutional Bundle for NGO/Hospitals

multi-user profiles, training, fleet monitoring,
support and replacement workflow

TIER 1

Optikensis Access

Software-only version that works with an
ordinary webcam.

- One-time software purchase
- No added hardware to start
- Core accessibility stays available
- Maintenance check: monthly/quarterly
diagnostics

TIER 2

MotionGlass Kit

IMU-based wearable for more precise
head-motion control.

- 6-axis IMU: accelerometer + gyroscope
- Bluetooth link to the Access Engine
- Works as glasses, clip-on or headband
module
- Optional Care plan for calibration and
support

THANK YOU

Hope to hear back from you guys , lets build
something **REVOLUTIONARY** together

